IST 691: Deep Learning in Practice
Final Project Proposal

NeuroVision: Deep Learning-Based Brain Tumor Detection and Classification from MRI Scans

Shane Rubenstein, Titus Kenneth George Kirubaharan, Brett Cerenzio, Christopher Marfisi

Project Overview:

Our project, NeuroVision, focuses on building a deep learning model that can learn how to automatically detect and classify brain tumors from MRI scans with a level of accuracy that can help in real clinical decision making. The goal of this project was not only to identify the presence of a tumor but also to correctly classify it into one of four categories such as glioma, meningioma, pituitary, or no tumor. We approached this problem using a convolutional neural network (CNN), supported by preprocessing steps and structured experiments and interpretability tools like Grad-CAM which will be very useful for us in finding exactly the region in the MRI that influenced the model's prediction. We used EfficientNet to parse through the MRI imaging which will help the project to blend with technical innovation for a healthcare impact. Overall, our main goal is to help doctors detect tumors sooner and with greater confidence. By reducing the need for time in consuming manual checks and increasing diagnostic accuracy. Our project can support radiologists and contribute better outcomes for patients

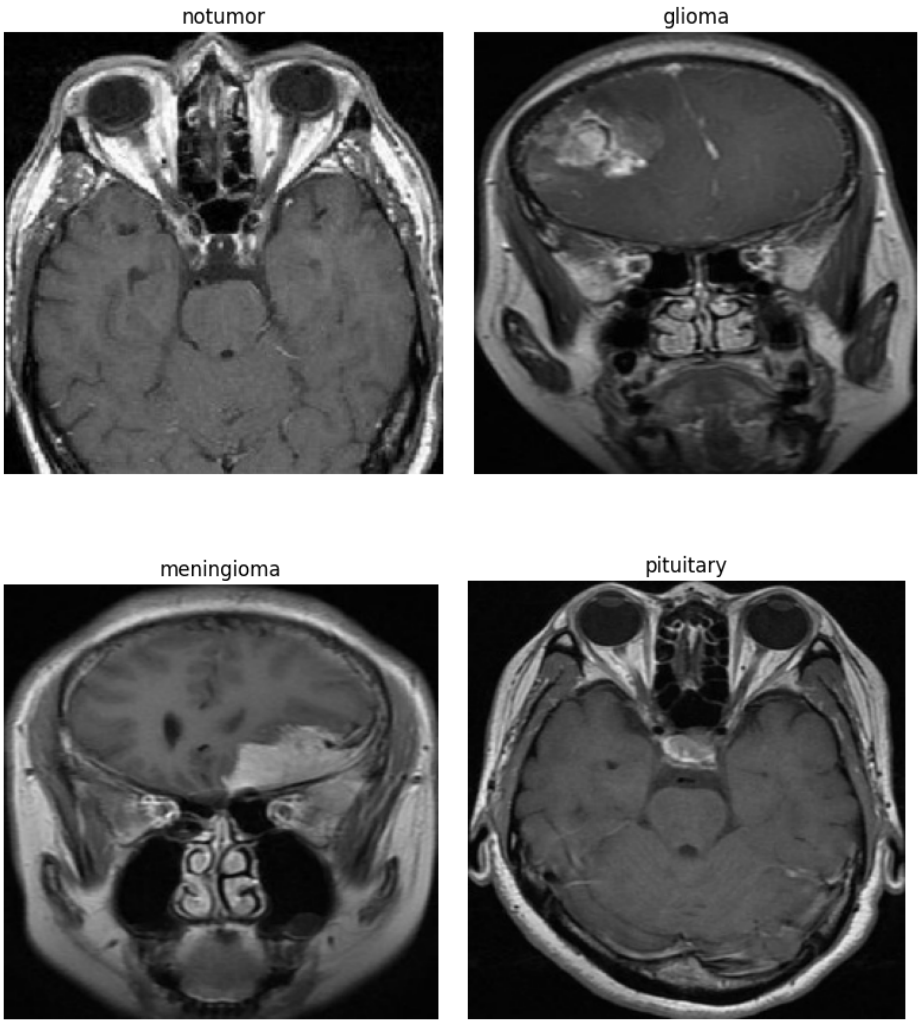
Prediction, Inference, and Other Goals:

The goal of this project was to predict the type of cancer someone had as well as if they had cancer or not to the highest degree possible. Initially we set some ambitious benchmarks in terms of levels of accuracy that we wanted to achieve. Comparing what was possible to other models, we saw that other models had a false negative rate of almost 0, so we wanted to achieve the same or even better. Conversely, we wanted our model to prioritize false negatives over false positives naturally as a false negative in the realm of cancer is much more serious than someone getting further inspection on a tumor that isn't there. In terms of overall accuracy within our model, we were looking for measurements over 99%, and ideally beating all the current models on the internet, as none of them had used a combination of a newer framework like EfficientNet as well as a combination of data sources. In the end, our main lofty goal was to produce a model with the highest accuracy of anything available.

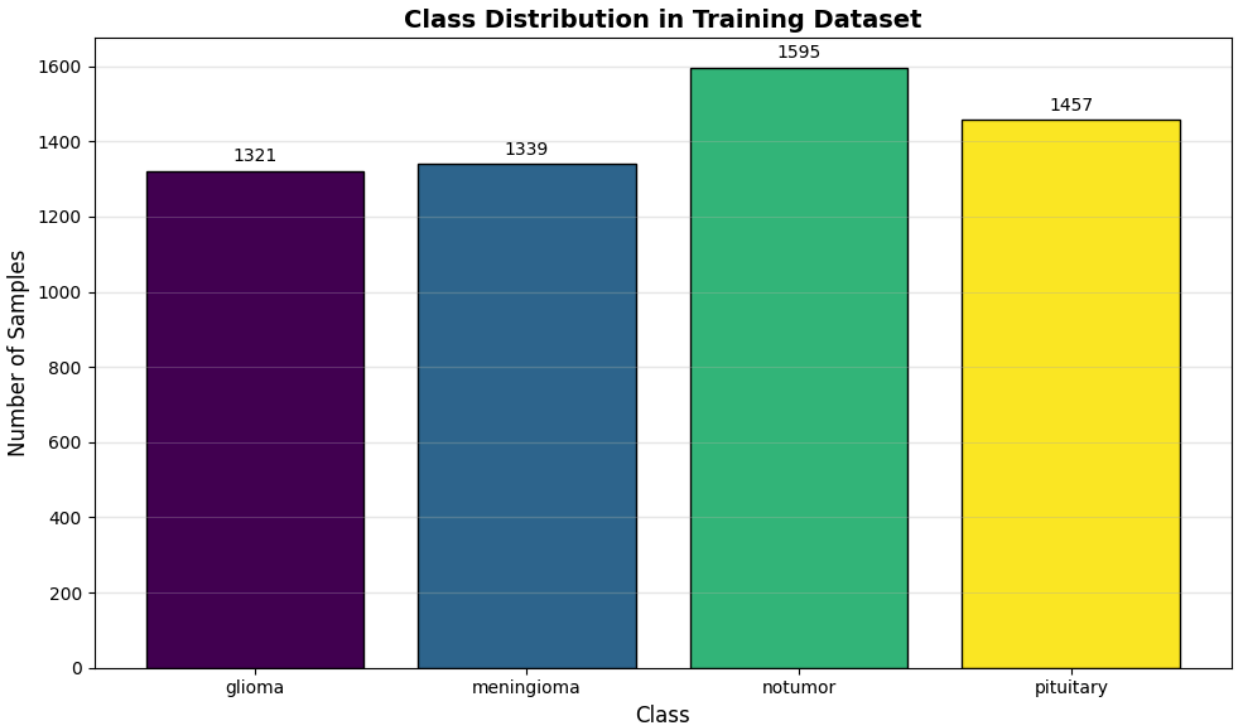
Data Exploration, Including Interesting Visualizations:

Given our data is entirely MRI scans of brains, there's not much initial exploration one can do aside from simply looking at the photos to identify what type of tumor indicates what type of brain cancer.

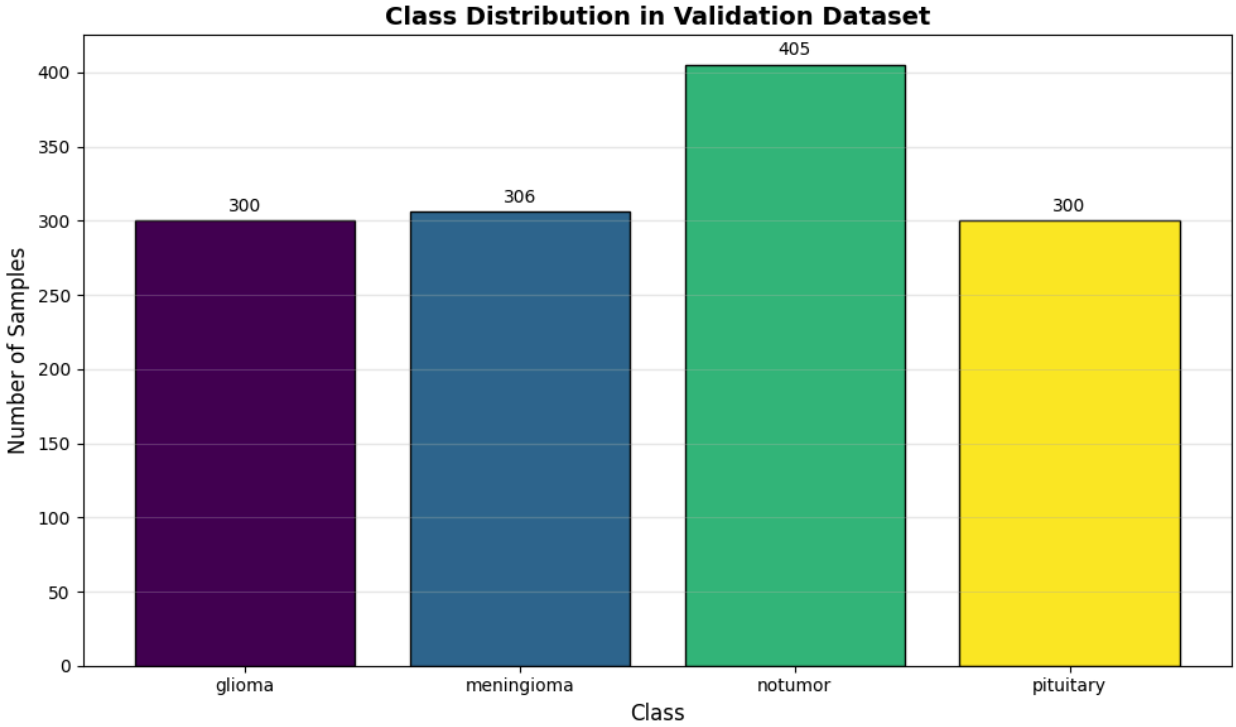
Below are images of types of brain cancer that we're analyzing as well as one with no tumor. Here you can see the general positions of the tumors relative to the type of cancer, where glioma is in the top left section of the brain , meningioma is in the bottom right section of the brain, and pituitary is in the central region of the brain.



Our training set consisted of the distribution below among the types of cancer. There are slightly more ‘no tumor’ observations than the other types, but this also aligns with real life, as most people who get MRIs of their brains do not have a tumor. The test set is significantly more populated towards no tumor, but since we’re training the model on a lower proportion of no tumor, no major bias is being instilled in the model.



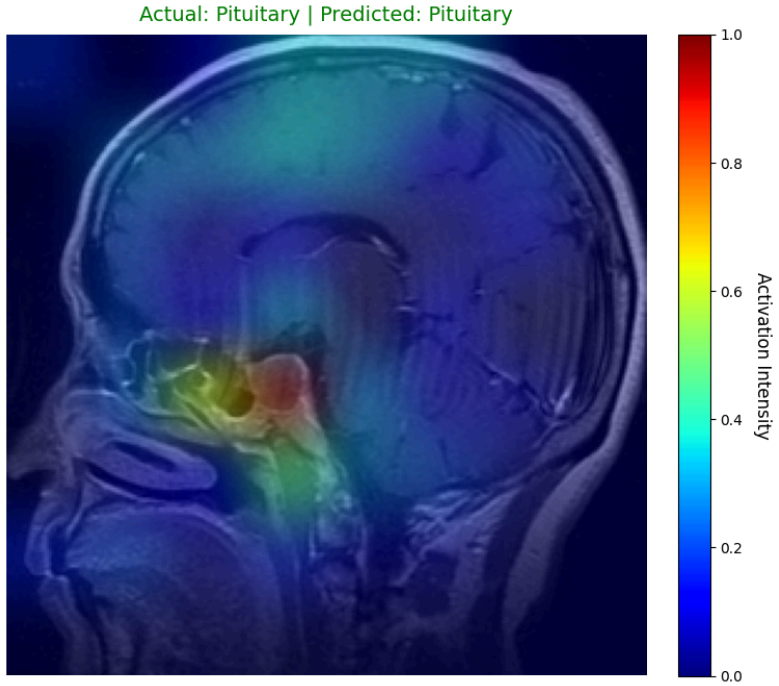
Here is the distribution of the test set as well.



Interesting/Surprising Results:

After running a couple versions of the model with different hyperparameters and different types of data preparation, we were able to surpass our expectations and produce an extremely accurate version of our model. There were in-fact 0 false positives or false negatives, and in terms of overall errors and misclassification, there were only 4 errors in all of the 1311 test observations. All of these errors were misclassifications of cancer types, the significantly less severe types of errors. As we got our model's total accuracy to 99.69%, we decided to look further if any other model had come close. There were models on the same dataset at 99.5% accuracy and 99.6%, but our model seemed to perform the best of all of them. There were some models that only classified the type of cancer, not looking into whether the person had cancer or not, which scored at 99.76%, but our model had more test observations and more classification areas as we included people with no tumors at all. The main surprising result was that we indeed outperformed every similar model available.

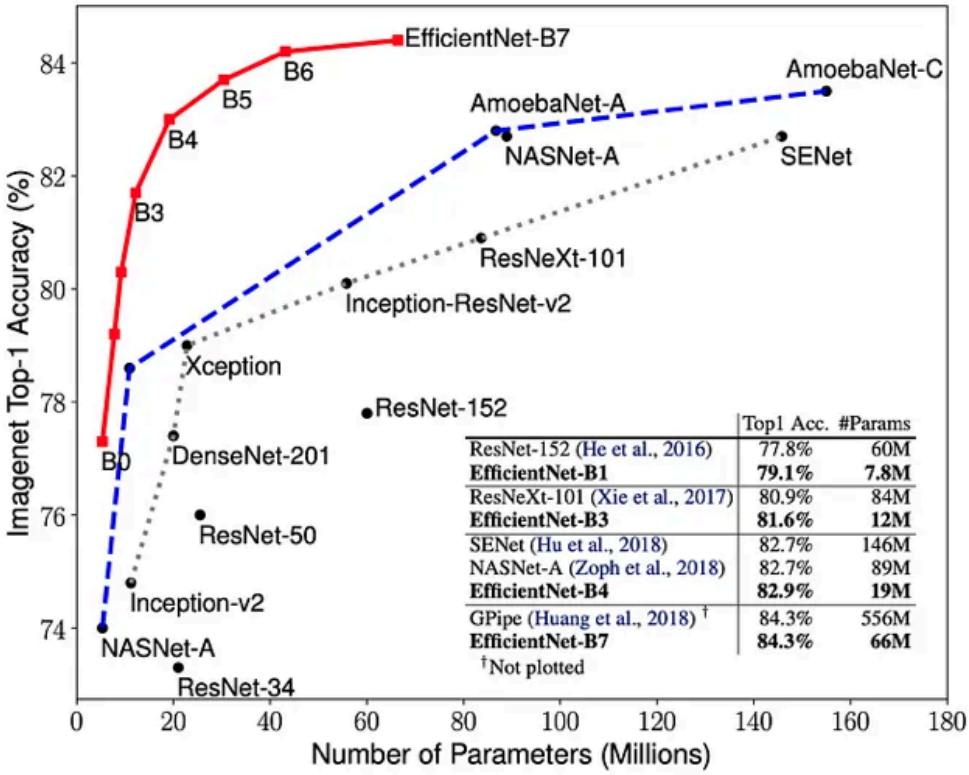
We also wanted to see what parts of the image that our model was focusing on to make its choices, so we utilized a gradCAM framework to show the model's focus. Below we can see where it's focusing when it is evaluating a pituitary gland tumor, with the red being the area of the most focus with a gradual fade on sections that it's not identifying.



Summary of Methods Used to Solve the Problem:

For this project, we decided to utilize the EfficientNetV2B0 model architecture pre-trained on the ImageNet dataset which is provided by Keras. The original author who put together the dataset utilized the ResNet50 model on the dataset, which achieved a test accuracy of 98.8% (Masoudnick). One kaggle user utilized the Xception model with some dropout layers

& an extra dense layer to achieve a test accuracy of 99.5% (Mohamed). We utilized the EfficientNetV2B0 model because its architecture allows it to have few parameters while maintaining a high level of accuracy that's comparable to Xception & ResNet50.



	Top1 Acc.	#Params
ResNet-152 (He et al., 2016)	77.8%	60M
EfficientNet-B1	**79.1%**	**7.8M**
ResNeXt-101 (Xie et al., 2017)	80.9%	84M
EfficientNet-B3	**81.6%**	**12M**
SENet (Hu et al., 2018)	82.7%	146M
NASNet-A (Zoph et al., 2018)	82.7%	89M
EfficientNet-B4	**82.9%**	**19M**
GPipe (Huang et al., 2018) †	84.3%	556M
EfficientNet-B7	**84.3%**	**66M**

†Not plotted

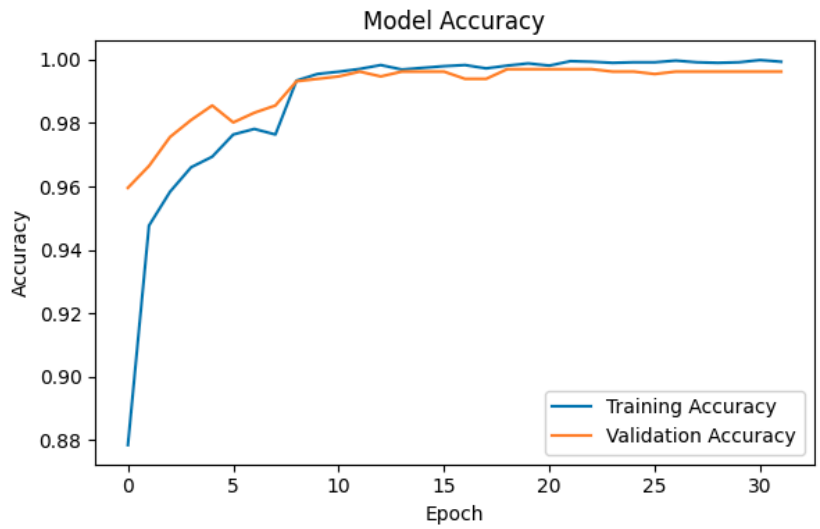
Source: arXiv:1905.11946

The graphic above shows how EfficientNet compares to ResNet50 & Xception in size & accuracy. This shows how the smallest EfficientNet models share similar accuracy scores as ResNet50 & Xception for nearly half of the total parameters. It does this by using what they call a compound scaling method, where the network uniformly scales the network width, depth, & resolution with a set of fixed compound scaling coefficients (Randellini). A smaller network allows our model to not only train quicker, but also make predictions quicker, which will aid in the efficiency of our model if it were to be used on a larger scale (such as in hospitals). We felt that we had enough images in our dataset that utilizing the smallest EfficientNet model (B0) would best fit our needs. Following Mohamed's framework, we added two dropout layers as well as a dense layer to the end of the EfficientNetB0 model to increase the predictive power of the model.

For image augmentation, we utilized pre-processing code from Masoudnick's github page which crops the blank space from the edges of the MRI photos. This ensured that there was no wasted space in the input images. We then added image transformations to the training & validation sets to make account for all MRI angles & image quality. For these reasons, I utilized a random horizontal flip to account for different MRI angles. Then, to account for image quality, I randomly added blurriness, oversaturation, & undersaturation to all the images in the dataset. This ensures that our model will be robust to all MRI images regardless of angle or poor quality.

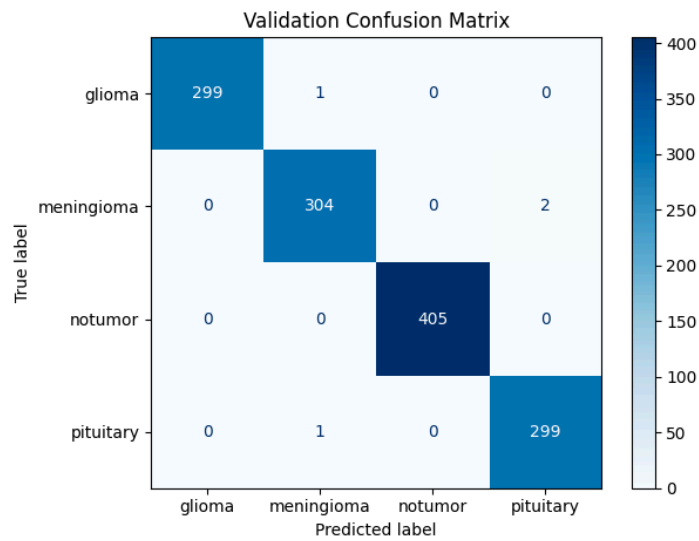
Results Summary:

As seen to the right, the final training accuracy of the model was 0.9995, and the final validation accuracy was 0.9969. Because of the robust dataset, proper image preprocessing, and thought-out model development, the model performed near-perfect without encountering overfitting.



Within the confusion matrix, there were only four misclassifications in the test dataset. There were no false positives for the "notumor" class, which is most important, as misclassifying a patient with cancer as someone without it could be life-altering. There were also no false negatives for the "notumor" class, which would save time and money of both hospitals and patients, as misclassifying a patient without cancer as someone with cancer could lead to unnecessary surgeries or extra tests.

```
Final Training Accuracy: 0.9995
Final Validation Accuracy: 0.9969
```




```
Classification Report:
              precision    recall  f1-score   support

      glioma       1.00      1.00      1.00       300
  meningioma       0.99      0.99      0.99       306
     notumor       1.00      1.00      1.00       405
   pituitary       0.99      1.00      1.00       300

    accuracy                           1.00      1311
   macro avg       1.00      1.00      1.00      1311
weighted avg       1.00      1.00      1.00      1311
```

Problems Encountered:

The biggest issues we ran into had to do with the decisions of what model to use and what image augmentation steps we wanted to take in order to ensure a robust model. EfficientNet has a lot of different model types that vary in size and accuracy and given the limited timespan, we knew we couldn't trial and error with each one to find the optimal model type. We ultimately utilized the B0 model to maximize model efficiency, and given the final results, I would say that it worked out well for us. However, in the future, it might be wise to try some of the bigger EfficientNet models such as the B3 or B4 models to see if that has any significant impact on model accuracy. For the image augmentation, there were many different augmentation types we could've looked into to alter the appearance of certain images. However, we didn't want to change the image too much, as MRIs images are taken in a very controlled environment by design, which is vastly different from other scenarios where image classification methods are used. For this reason, we opted for a more simplistic image augmentation process.

Summary of How Well you Achieved Your Prediction and Inference Goals:

The goals of our project were significantly surpassed. While the team intended to create a model that outperformed existing models at classifying a singular class, the final model outperformed existing models at making any classification. The validation accuracy of 99.69% was greater than any model currently displayed online in research papers and Kaggle notebooks. The current leading model has a validation accuracy of 99.54%, and with cancer being so lethal, this improvement of 0.15% could save lives.

Citations:

Jyotismita Chaki, "Brain Tumor MRI Dataset", IEEE Dataport, July 26, 2023, doi:10.21227/1jny-g144

Masoudnick, *Brain-Tumor-MRI-Classification.* Github, 2021, https://github.com/masoudnick/Brain-Tumor-MRI-Classification/

Mohamed, Yousef. "Brain Tumor MRI 99% Accuracy." Kaggle, 2023, https://www.kaggle.com/code/yousefmohamed20/brain-tumor-mri-accuracy-99

Randellini, Enrico. "Image Classification: ResMet vs EfficientNet vs EfficientNet_v2 vs Compact Convolutional Transformers." *Medium*, 5 Jan. 2023, medium.com/@enrico.randellini/image-classification-resnet-vs-efficientnet-vs-efficientnet-v2-vs-compact-convolutional-c205838bbf49.